

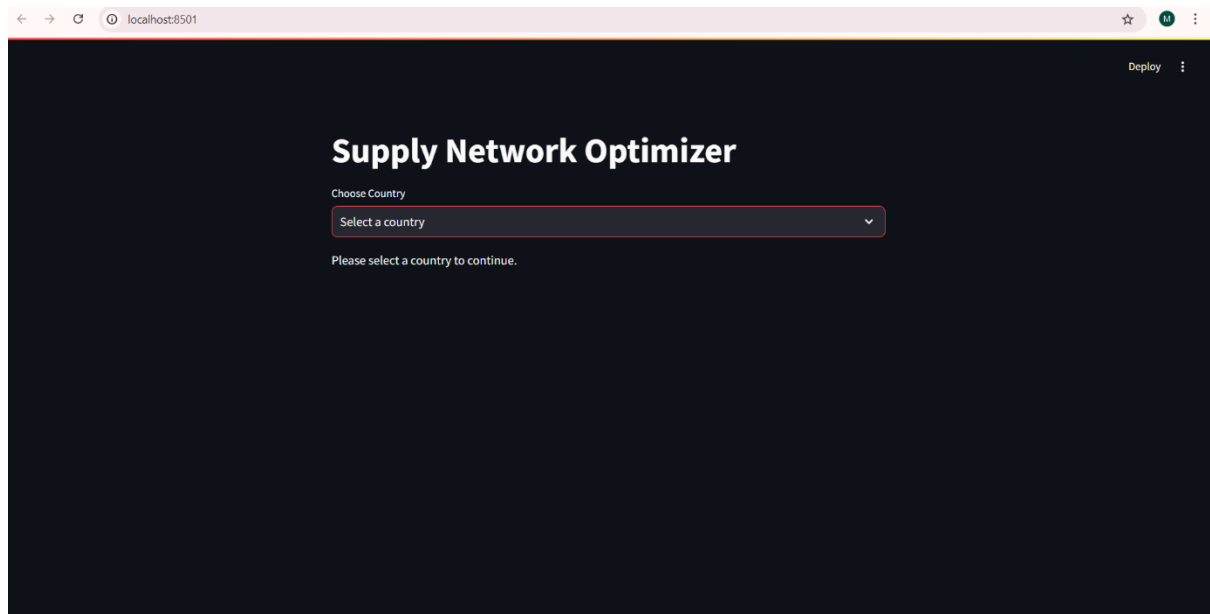
User Manual: Supply Network Optimizer

1. Introduction

The application is designed to optimize a supply network by analyzing distributor sales data, predicting future sales, and visualizing distances between manufacturing plants and distributors on a map.

2. Features

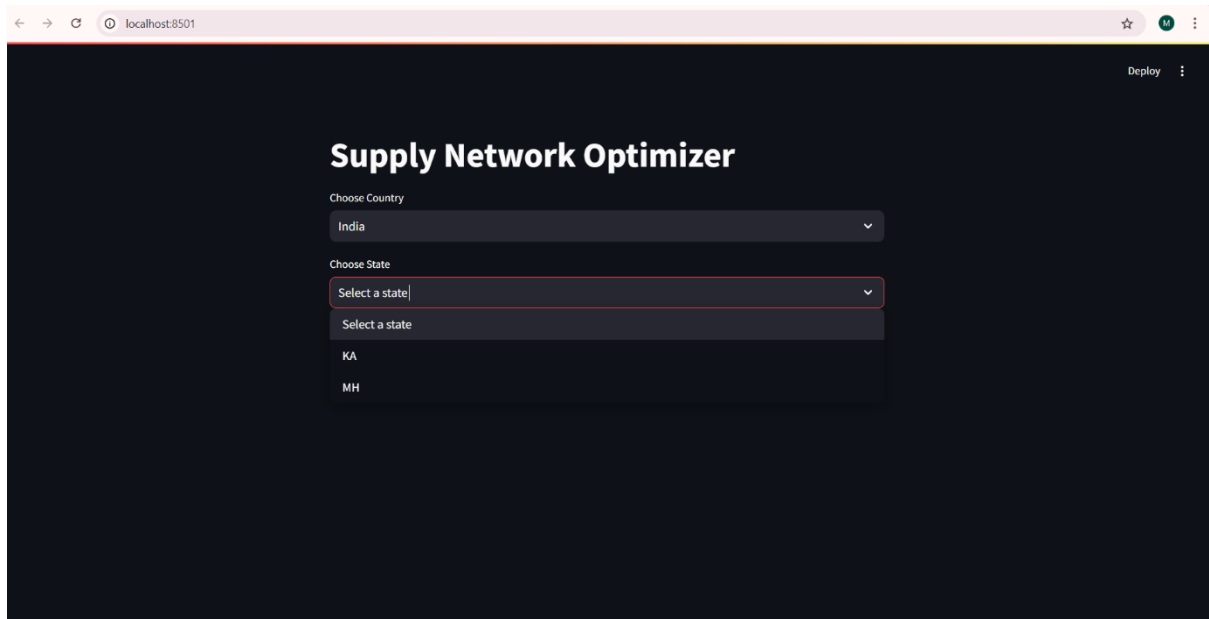
- **User Selections:** The user can select a country, state, city, category, and month, dynamically filtering the data based on their selections.
- **Sales Prediction:** The application calculates predicted sales based on historical data for selected distributors and displays this information.
- **Map Visualization:** A map is generated using Folium, showing manufacturing plants and distributors, along with distance calculations.
- **Simulation:** A button triggers a simulation that may include additional calculations or visualizations.



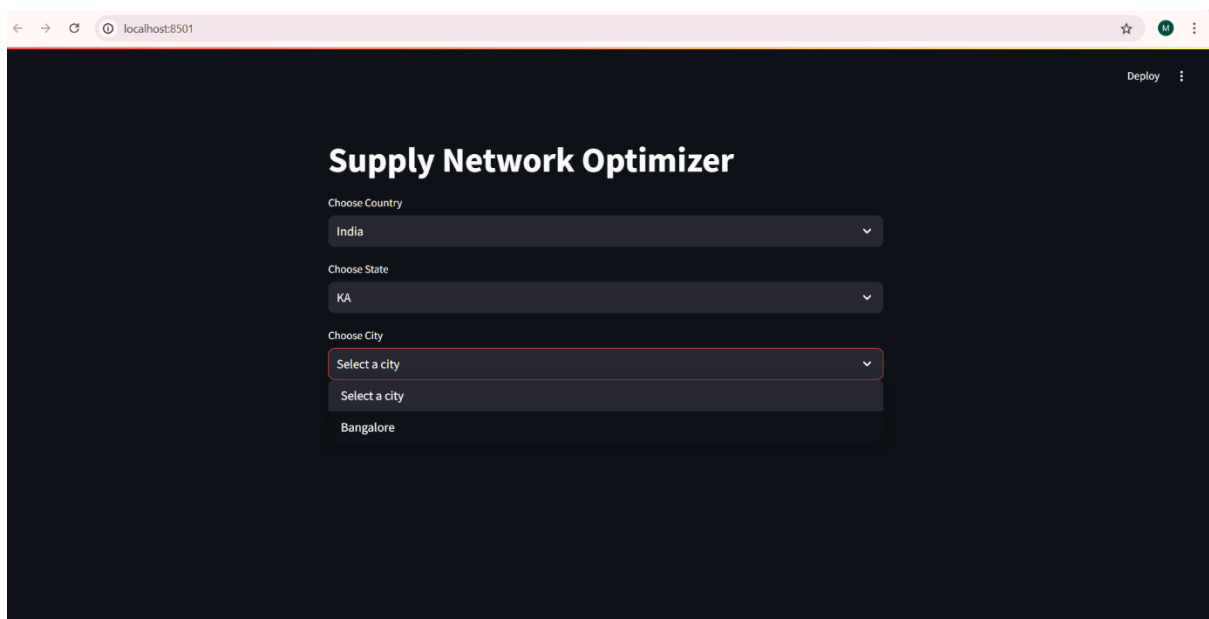
3. Steps to Use the Supply Network Optimizer

Step 2: Access the Application

- The user should select a country, state, city, category and month.



The screenshot shows a web browser window at localhost:8501 displaying the 'Supply Network Optimizer' application. The interface has a dark theme. At the top right, there is a 'Deploy' button and a menu icon. The main content area features the title 'Supply Network Optimizer' in a large, bold font. Below the title, there are two selection steps. The first step, 'Choose Country', has a dropdown menu with 'India' selected. The second step, 'Choose State', has a dropdown menu with 'Select a state' selected. Below the 'Choose State' dropdown, there is a list of states: 'KA' and 'MH'.



The screenshot shows the same web browser window at localhost:8501, but now the 'Choose City' dropdown menu is open, showing 'Select a city' and 'Bangalore' as options. The 'Choose State' dropdown menu remains open, showing 'Select a state' and 'KA' as options. The 'Choose Country' dropdown menu remains open, showing 'India' as the selected option.

localhost:8501

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Deploy

Supply Network Optimizer

Choose Country

India

Choose State

KA

Choose City

Bangalore

Choose Category

Select a category

Select a category

Painkiller

Allergy

Antibiotic

localhost:8501

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Deploy

Supply Network Optimizer

Choose Country

India

Apr

May

Jun

Jul

Aug

Sep

Oct

Jan

Predicted Sales

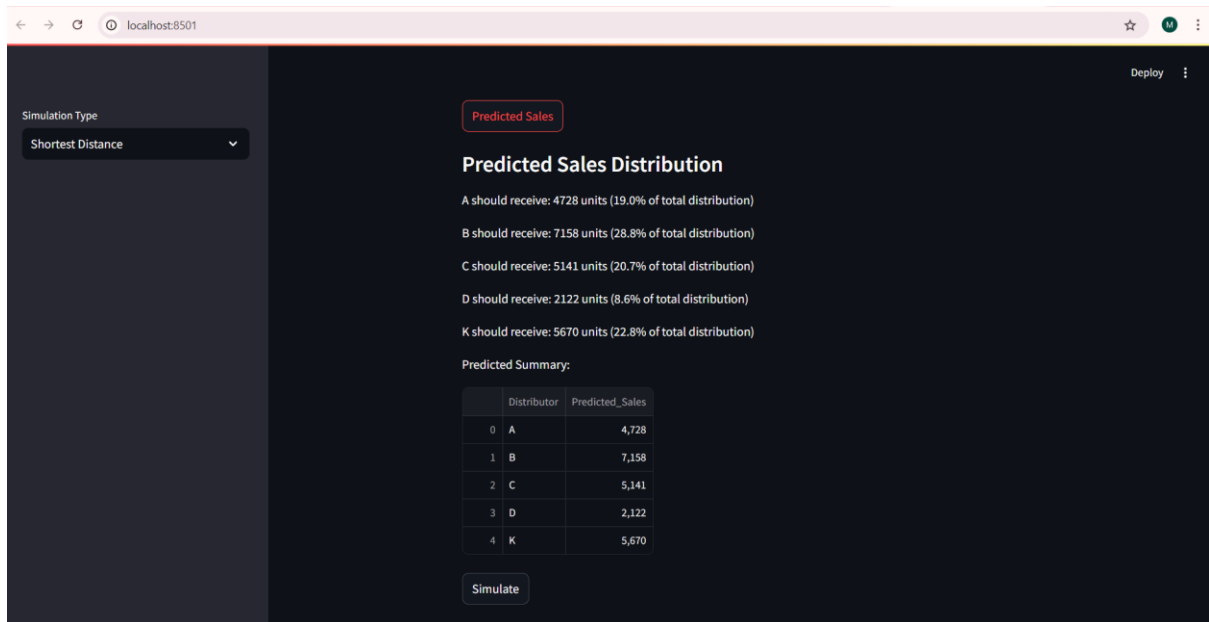
Step 2: Predicted Sales

- Click the **Predicted Sales** button. The supply network optimizer will automatically give Predicted Summary with Distributors name and Predicted_Sales.

The screenshot shows a web application titled "Supply Network Optimizer" running on localhost:8501. The interface has a dark theme. On the right side, there are five dropdown menus for selecting parameters: "Choose Country" (set to India), "Choose State" (set to KA), "Choose City" (set to Bangalore), "Choose Category" (set to Painkiller), and "Choose Month" (set to Jan). Below these dropdowns is a red button labeled "Predicted Sales". In the top right corner, there is a "Deploy" button and a menu icon.

This screenshot shows the same web application after clicking the "Predicted Sales" button. A new section titled "Predicted Sales Distribution" has appeared below the input fields. On the left side, a new "Simulation Type" dropdown menu is visible, set to "Shortest Distance". The "Predicted Sales" button remains visible. The distribution results are listed as follows:

Distributor	Units	Percentage of Total Distribution
A	4728	19.0%
B	7158	28.8%
C	5141	20.7%
D	2122	8.6%
K	5670	22.8%



Step 3: Simulate

- Click the **Simulate** button. The supply network optimizer will give Distances from Manufacturing_Plant to distributors with Distributors name and Distance.
- It will give Best route From Manufacturing_Plants.
- The best route from the manufacturing plants to the distributors is clearly visible.

